Pebble Firmware Version 4.00e
Operating Manual

Document Revision Date: Nov 2, 2026



Introduction

This is the operating manual for the firmware version 4.00e that can be installed on the Pebble HF radio. The Pebble 4.00e firmware was derived from the 4.00e uSDX firmware by Rob (GW8RDI). Since the pebble is a simplified uSDX, it was possible to simplify the firmware which freed up some program space for a few new features.

NOTE: This is experimental firmware and it is not endorsed by or supported by the Pebble HF project developers.

User Interface (UI) Controls

Pebble HF has three UI controls:

- **Tuning Knob (Rotary Encoder)** — The main knob on the right side. Turn to tune frequency. Press and turn to adjust volume. Press to change tuning step size.
- **MENU Button (Left)** — Short press enters the menu system. Used to navigate and select menu parameters.
- **SELECT Button (Right)** — Used to exit menus, cycle filter bandwidth (double-click), and toggle VFO/RIT (long press).

Basic Operation

Tuning

Turn the rotary encoder to change frequency.

Tuning Step Size

A short press the encoder to increase the tuning step size. Long press the encoder to decrease it. Available step sizes: 10M, 1M, 0.5M, 100k, 10k, 1k, 0.5k, 100, 10, 1 Hz.

Volume

Press and hold the encoder while turning to adjust volume (0–16).

Filter Bandwidth

Double-click the SELECT (right) button to cycle through receiver filter bandwidths. In SSB mode: Full, 3000, 2400, 1800 Hz. In CW mode: 500, 200, 100, 50 Hz. The filter also controls SSB transmit bandwidth.

VFO A/B and RIT

A long press the SELECT (right) button to toggle RIT (Receive Incremental Tuning). A second long press swaps between VFO A and VFO B.

Factory Reset

Hold any UI button down during power-up to reset all settings to factory defaults.

The Menu System

Press the MENU (left) button to enter the menu. Turn the encoder to scroll through menu items. Press the MENU button again to select a parameter for editing, then turn the encoder to change the value. Press the SELECT (right) button to save and go back, or press it again to exit the menu entirely.

Quick menu access: Hold the MENU (left) button while turning the encoder to quickly navigate to a parameter. Release the button and turn the encoder to immediately change the value.

Menu Reference — Summary

#	Menu Item	Values	Description
1.1	Volume	1 to 16	Radio volume
1.2	Tone Volume	1 to 16	CW side-tone volume
1.3	Mode	LSB, USB, CW	Modulation mode
1.4	Filter BW	Full, 3000, 2400, 1800, 500, 200, 100, 50 Hz	Receiver/transmitter audio passband filter
1.5	VFO Mode	A, B	Select VFO A or VFO B
1.6	RIT	OFF, ON	Receive Incremental Tuning

#	Menu Item	Values	Description
1.7	AGC	OFF, ON	Automatic Gain Control
1.8	NR	0–8	Noise reduction level
1.9	ATT	0, -13, -20, -33, -40, -53, -60, -73 dB	Analog front-end attenuator
1.10	ATT2	0–16 (6 dB steps)	Digital attenuator in CIC stage
1.11	S-meter	OFF, dBm, S, S-bar, wpm	Signal meter display format
2.1	CW Decoder	OFF, ON	On-screen CW decoder
2.2	CW Tone	600Hz or 700Hz	CW sidetone frequency
2.3	CW Interval	1 to 60	Number of seconds to wait before repeating a CW message
2.4	Semi QSK	OFF, ON	Mutes RX briefly after keying for semi break-in
2.5	Keyer Speed	1–60 WPM	CW keyer speed in words per minute
2.6	Keyer Mode	Iambic A, Iambic B, Ultimatic, Straight	Type of CW keyer
2.7	Keyer Swap	OFF, ON	Swap DIT/DAH paddle inputs
2.8	Practice	OFF, ON	Disables transmit for CW practice
3.1	VOX	OFF, ON	Voice-operated transmit
3.2	Noise Gate	0–255	Audio threshold for VOX and SSB TX
3.3	TX Drive	0–8	Transmit audio gain (6 dB steps)
3.4	PA Bias Min	0-255	PA PWM min level
3.5	PA Bias max	0–255	PA PWM max level
3.6	Ref freq	26000000 typical	SI5351 crystal calibration
3.7	IQ Phase	0–180°	RX I/Q phase correction
3.8	Quad	OFF, ON	Some DSP magic

4.1	Freq Presets	Sub Menu	Store and Recall up to 8 frequency/mode values
4.2	CW Messages	Sub Menu	Store and Playback up to 6 CW messages
4.3	User Info	Sub Menu	User Name and Call-Sign
4.4	Firmware Ver	(read only)	Firmware version and compile date

---

Menu Reference — Detailed

1.1 Volume

Values: -1 through 16

Controls the main audio output level of the receiver. Setting 1 is very quiet; 16 is maximum volume. Volume can also be adjusted at any time by pressing and holding the encoder knob while turning it.

1.2 Tone Volume

Values: -1 through 16

Sets the CW sidetone volume independently from the main receiver volume. This allows you to have a comfortable sidetone level while keeping the receiver volume at a different level. Default is 12.

1.3 Mode

Values: LSB, USB, CW

Selects the modulation mode.

- **LSB** — Lower Sideband
- **USB** — Upper Sideband
- **CW** — Continuous Wave (morse code). When CW mode is selected, the filter automatically narrows and the tuning step size adjusts for CW operation.

1.4 Filter BW

Values: Full, 3000, 2400, 1800, 500, 200, 100, 50 Hz

Sets the DSP audio pass band filter width for both receive and transmit.

- **Full** — No filtering, full audio bandwidth.
- **3000 Hz** — Wide SSB filter (300–3000 Hz passband).
- **2400 Hz** — Standard SSB filter (300–2400 Hz passband).
- **1800 Hz** — Narrow SSB filter (300–1800 Hz passband).
- **500 Hz** — Wide CW filter, good for general CW operation.
- **200 Hz** — Medium CW filter, helps isolate a single signal.
- **100 Hz** — Narrow CW filter for crowded band conditions.
- **50 Hz** — Ultra-narrow CW filter for extreme selectivity.

Quick access: Double-click the SELECT (right) button to cycle through filters without entering the menu. In SSB mode, it cycles through Full/3000/2400/1800. In CW mode, it cycles through 500/200/100/50.

1.5 VFO Mode

Values: A, B

Selects which VFO (Variable Frequency Oscillator) memory is active. Each VFO stores its own frequency and mode. This allows you to quickly switch between two frequencies.

Quick access: Long press the SELECT (right) button to toggle RIT, then long press again to swap VFOs.

1.6 RIT

Values: OFF, ON

Receive Incremental Tuning allows you to offset the receive frequency from the transmit frequency. When RIT is enabled, the tuning step automatically changes to 10 Hz for fine adjustment. This is useful for tuning in a station that is slightly off your transmit frequency without moving your transmit frequency.

Quick access: Long press the SELECT (right) button to toggle RIT on or off.

1.7 AGC

Values: OFF, ON

Automatic Gain Control automatically adjusts the receiver volume to keep signals at a consistent level — boosting weak signals and reducing strong ones. AGC is disabled by default.

- **OFF** — Manual volume control only. Quieter but requires more manual adjustment.
- **ON** — Automatic level control. Good for SSB but can be distracting during CW.

1.8 NR

Values: 0–8

Noise Reduction applies a low-pass smoothing filter to reduce background noise. Higher values apply more aggressive noise reduction. A setting of 0 disables noise reduction. Start with low values (1–3) and increase as needed. NR is reset to 0 when switching to CW mode.

1.9 ATT

Values: 0dB, -13dB, -20dB, -33dB, -40dB, -53dB, -60dB, -73dB

Analog front-end attenuator. Reduces the signal level entering the receiver. Use this when strong signals are overloading the receiver or to reduce atmospheric noise. On the 20-meter band, a small amount of attenuation (e.g., -13dB) can improve reception by lowering the noise floor relative to signals.

1.10 ATT2

Values: 0–16 (each step is approximately 6 dB)

Digital attenuator applied in the CIC (Cascaded Integrator-Comb) filter stage of the DSP receiver. Provides additional signal level control in the digital domain. Use in combination with ATT for fine-tuning the receiver's dynamic range.

1.11 S-meter

Values: OFF, dBm, S, S-bar, wpm

Controls what is displayed on the signal meter area of the LCD.

- **OFF** — No signal meter displayed.
- **dBm** — Shows received signal strength in dBm.
- **S** — Shows signal strength as an S-unit value (S1–S9, S9+).
- **S-bar** — Shows signal strength as a visual bar graph.
- **wpm** — Shows decoded CW speed in words per minute (useful with CW Decoder enabled).

2.1 CW Decoder

Values: OFF, ON

Enables or disables the built-in CW decoder. When enabled, received CW signals are decoded and the characters are displayed on the LCD screen. Works best with a narrow CW filter (200 Hz or less) and a clean signal.

2.2 CW Tone

Values: 600Hz or 700 Hz

Sets the CW sidetone frequency in Hertz. This is the pitch of the tone you hear when keying CW, and it also sets the CW offset used for transmit. Choose a pitch that is comfortable for your ear. Default is 600 Hz.

2.3 CW Interval

Values: 1 to 60 seconds

Sets the delay between automated CQ calls (stored messages)

2.4 Semi QSK

Values: OFF, ON

Semi QSK (semi break-in) mutes the receiver for a short period after each CW element (dit or dah) and during word spaces. This prevents you from hearing your own transmitted signal between elements while still allowing you to hear the band during longer pauses. Useful for comfortable CW ragchewing.

2.5 Keyer Speed

Values: 1–60 WPM

Sets the speed of the built-in CW keyer in words per minute (Paris standard). This setting applies to Iambic A and Iambic B modes. In Straight key mode, speed is determined by your manual keying. Default is 15 WPM.

2.6 Keyer Mode

Values: Iambic A, Iambic B, Ultimatic, Straight

Selects the type of CW keyer.

- **Iambic A** — Iambic keyer mode A. Alternating dits and dahs are produced when both paddles are squeezed. Releasing the paddles stops keying immediately.
- **Iambic B** — Iambic keyer mode B. Similar to Iambic A, but an additional dit or dah is inserted after releasing the paddles if the opposite paddle was held.
- **Ultimatic** — Instead of alternating dits and dahs like an Iambic keyer, Ultimatic mode gives priority to the most recently pressed paddle.
- **Straight** — Straight key mode. The key input is used as a simple on/off switch. This is the default and is the correct mode for the built-in key button.

> **Important:** If you select Iambic A or Iambic B mode, you **must** have a paddle connected to the key/mic jack. If nothing is plugged in, the built-in microphone can falsely trigger the keyer and cause unintended transmissions.

2.7 Keyer Swap

Values: OFF, ON

Swaps the DIT and DAH paddle inputs. Use this if your paddle is wired in reverse, or if you prefer to key left-handed. OFF is normal (tip = DIT, ring = DAH). ON swaps them.

2.8 Practice

Values: OFF, ON

When enabled, the transmitter is disabled for CW practice purposes. You will hear the sidetone in your speaker but no RF is transmitted. This is useful for practicing your keying without going on the air.

3.1 VOX

Values: OFF, ON

Voice-Operated Transmit automatically switches the radio to transmit when audio is detected from the microphone. The sensitivity is controlled by the Noise Gate setting (3.2). VOX is also used for digital mode operation — the audio from your computer triggers the transmitter.

When VOX is enabled, a "V" appears on the right side of the display instead of "R" (receive).

> **Note:** VOX is always disabled on power-up for safety, regardless of the saved setting.

3.2 Noise Gate

Values: 0–255

Sets the audio threshold level for VOX activation and SSB transmit. Audio below this level will not trigger transmission. Higher values require louder audio to trigger transmit. Lower values make VOX more sensitive.

- For VOX voice operation, start around 40–50 and adjust based on your environment.
- For digital modes, adjust so that your computer's audio reliably triggers transmit without false triggers from noise.

Default is 50.

3.3 TX Drive

Values: 0–8

Controls the transmit audio gain in approximately 6 dB steps. Higher values increase the modulation depth and transmit power.

- **0** — Minimum drive.
- **8** — Maximum / constant amplitude (useful for CW or digital modes).
- **2–4** — Typical range for SSB voice.

Adjust while monitoring your signal or based on signal reports. Default is 4 (set during initialization).

3.4 PA Bias min

Values: 0 to (PA Bias max - 1)

Sets the PWM (Pulse Width Modulation) level that represents 0% RF output from the power amplifier. This is a calibration parameter. **Do not change this unless you understand PA biasing.** Incorrect values can damage the PA transistors.

3.5 PA Bias max

Values: (PA Bias min) to 255

Sets the PWM level that represents 100% RF output from the power amplifier. This is a calibration parameter. **Do not change this unless you understand PA biasing.** Default is 128.

3.6 Ref freq

Values: typical 26000000

The actual crystal oscillator frequency of the SI5351 PLL. Used for frequency calibration. The Pebble HF uses a 26 MHz TCXO, so this should be close to 26000000. If your displayed frequency does not match a known reference signal, adjust this value until it does.

3.7 IQ Phase

Values: 0–180°

Adjusts the I/Q phase offset for the SDR receiver in degrees. This calibration parameter affects sideband suppression. The optimal value cancels out the

unwanted sideband. Adjust while listening to a signal and minimize the opposite sideband bleed-through.

3.8 Quad

Values: OFF, ON

Invert CLK2 in case of a huge phase-change

4.1 Freq Presets

This is a sub-menu for storing and recalling up to 8 frequency/mode pairs

---

4.2 CW Messages

This is a sub-menu for storing and playing back up to 6 CW messages

---

4.3 User Info

This is a sub-menu for storing the user name and call-sign which are used in CW messages

---

4.4 Version

Displays the firmware version number and compile date.

Button Quick Reference

Action	How	Context
Tune frequency	Turn encoder	Main screen
Adjust volume	Press + turn encoder	Any time
Change tuning step (up)	Short press encoder	Main screen
Change tuning step (down)	Long press encoder	Main screen
Cycle filter bandwidth	Double-click SELECT (right)	Main screen
Toggle RIT / Swap VFO	Long press SELECT (right)	Main screen
Enter menu	Short press MENU (left)	Main screen
Quick menu access	Hold MENU (left) + turn encoder	Main screen
Select menu item	Short press MENU (left)	In menu
Save and exit to menu	Short press SELECT (right)	Editing value
Exit menu	Short press SELECT (right)	In menu
Factory reset	Push any button during power-up	Power-up

Credits

- **Original uSDX Firmware:** Guido PE1NNZ
- **4.00e uSDX Firmware:** Rob GW8RDI
- **4.00e Pebble Firmware:** Scott KJ7NLA

Note: There were many more contributors to the open source uSDX project. Apologies for not being able to list them all here.